Response to the Reviewer: Failure-Case Analysis

Reviewer comment: *Notably, one of the recommendations is to include failure-case analysis for unsuccessful interception and delayed cooperative engagement scenarios because understanding framework limitations remains essential for practical battlefield deployment reliability.*

Response: Thank you for this important recommendation. We have added a dedicated failure-case analysis that identifies the terminal miss mechanism, quantifies the resulting cooperative-engagement delay, and examines the roles of sensing and actuation imperfections. The analysis uses one reproducible boundary case from each engagement scenario. These cases are intentionally non-catastrophic: all eight attacking vehicles are neutralized through assignment redundancy, but one assigned interceptor fails to enter the 3 m hit region. They therefore expose a practically relevant limitation that can be hidden by target-level success statistics alone.

Boundary-failure outcomes. The two examples were obtained with the trained policy fixed under a one-sample (0.05 s) observation delay, position and velocity measurement-noise standard deviations of 3 m and 0.3 m/s, respectively, and the nominal command dynamics. The evaluation horizon was 75 s, and the cooperative tolerance was 0.5 s. The reported runs satisfy 8/8 target coverage, 19/20 individual interceptor hits, and 7/8 complete assignment groups.

Table 1: Observed boundary failures. The miss margin is $d_{\min} - d_{\text{hit}}$, where $d_{\text{hit}} = 3$ m.

Case	Seed	Hits	Coverage	Incomplete group	t_{CPA} (s)	$d_{\min}$ (m)	Margin (m)	Delay lower bound (s)
1	76048	19/20	8/8	A_6	33.10	3.243	0.243	> 41.90
2	77008	19/20	8/8	A_2	26.15	3.574	0.574	> 48.90

Direct failure mechanism. The incomplete groups arise from different interceptor–target pairs: $D_6 \rightarrow A_6$ in Case 1 and $D_2 \rightarrow A_2$ in Case 2. Both interceptors reach the terminal neighborhood but pass outside the hit sphere. In Case 1, the D_6–A_6 range decreases at 77.15 m/s one second before closest approach, reaches 3.243 m at 33.10 s, and then begins increasing; the range-rate sign change is observed one integration sample later. In Case 2, the corresponding values for D_2–A_2 are 79.22 m/s, 3.574 m, and 26.15 s. Thus, the immediate cause is a closest-approach point outside the terminal hit region, followed by separation, rather than failure to approach the assigned target.

The terminal geometry is demanding. At closest approach, the line-of-sight (LOS) angular rates are 23.89 rad/s and 19.31 rad/s, and the velocity-to-LOS errors are 97.3° and 77.9° for Cases 1 and 2, respectively. However, neither lateral command component saturates during the final two seconds; the maximum command-vector magnitudes are 0.648 g and 0.789 g. Consequently, these failures are better characterized as terminal phase/geometry errors induced by imperfect state information and command dynamics, not as insufficient control authority. Case 2 is additionally exposed to a target acceleration of up to 6.80 m/s^2 during the final two seconds, whereas the Case 1 target is nearly non-maneuvering in the same terminal window. This explains the larger miss margin and heading error in Case 2.

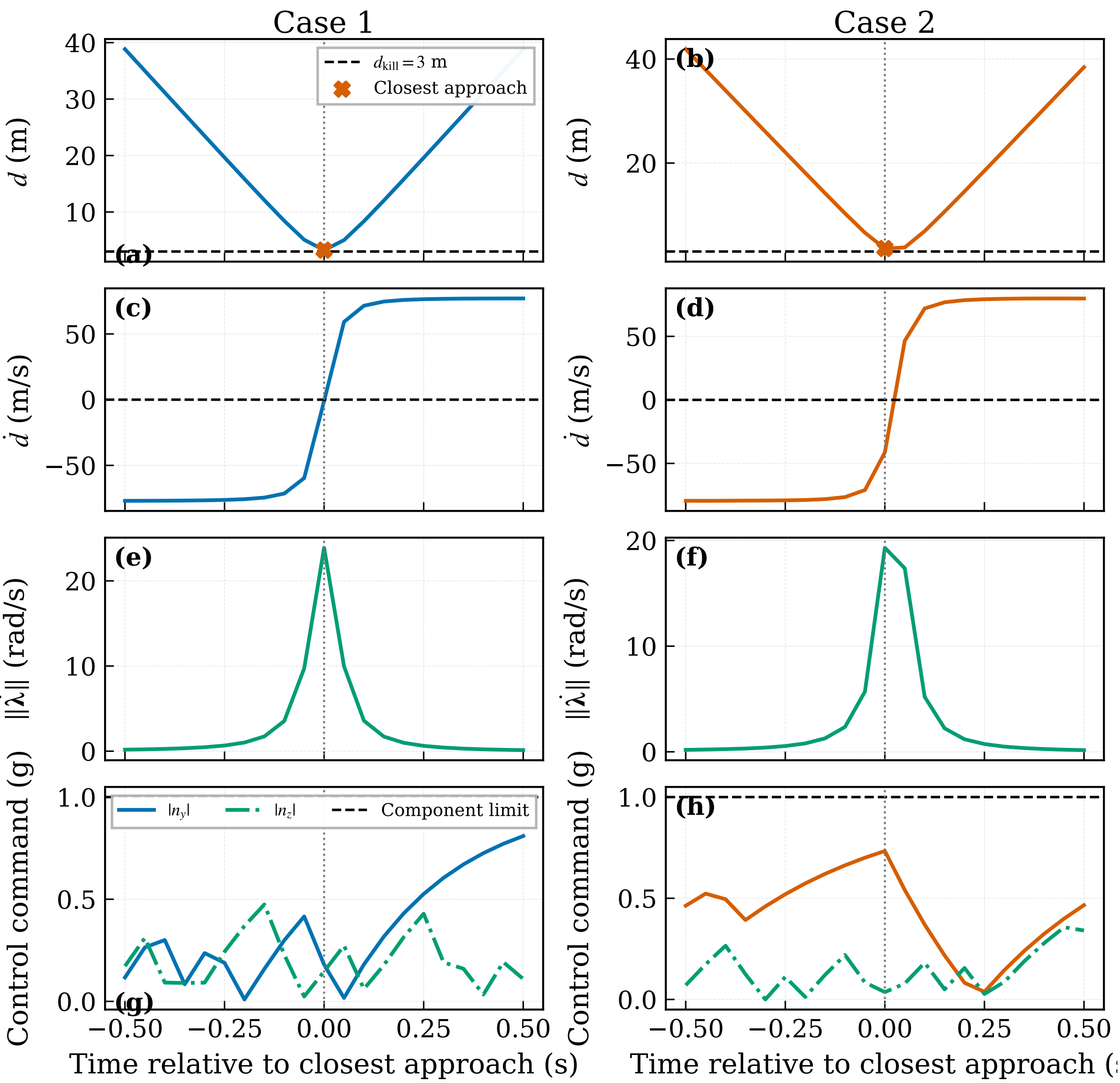

Figure 1: Terminal-window diagnosis for the missed D_6–A_6 (Case 1) and D_2–A_2 (Case 2) engagements. The distance minimum remains above the 3 m hit radius in both cases. The subsequent change from negative to positive $\dot{d}$ confirms passage through closest approach and transition to separation. The LOS-rate spike reveals the severe terminal geometry, while $|n_y|$ and $|n_z|$ remain below their component limits.

Why the cooperative engagement is delayed. In Case 1, D_6 and D_{14} are assigned to A_6. D_{14} intercepts A_6 at 33.10 s, exactly when D_6 reaches its 3.243 m closest point. In Case 2, the redundant D_2, D_{10}, D_{18} group is assigned to A_2. D_{18} and D_{10} intercept A_2 at 26.10 s and 26.15 s, while D_2 passes 0.574 m outside the hit boundary at 26.15 s. All other complete groups satisfy the 0.5 s tolerance; their maximum hit-time spreads are 0.05 s in Case 1 and 0.35 s in Case 2.

Because the respective missed member never generates a hit event, strict all-member group completion is right-censored at the 75 s horizon. Measured from the first peer hit, the cooperative-completion delay is therefore greater than 41.90 s in Case 1 and 48.90 s in Case 2. This distinction is important: the mission-level threat is removed on time, but the stricter assigned-group objective fails because one member does not recover after closest-approach passage.

Case 2 also contains a different type of delay in the otherwise complete A_3 group. Its three members hit at 64.55, 64.80, and 64.90 s. Hence A_3 is intercepted much later in absolute time than the other complete groups, but its within-group spread is only 0.35 s < 0.5 s. It is therefore a *delayed but synchronized* cooperative engagement, not a synchronization failure.

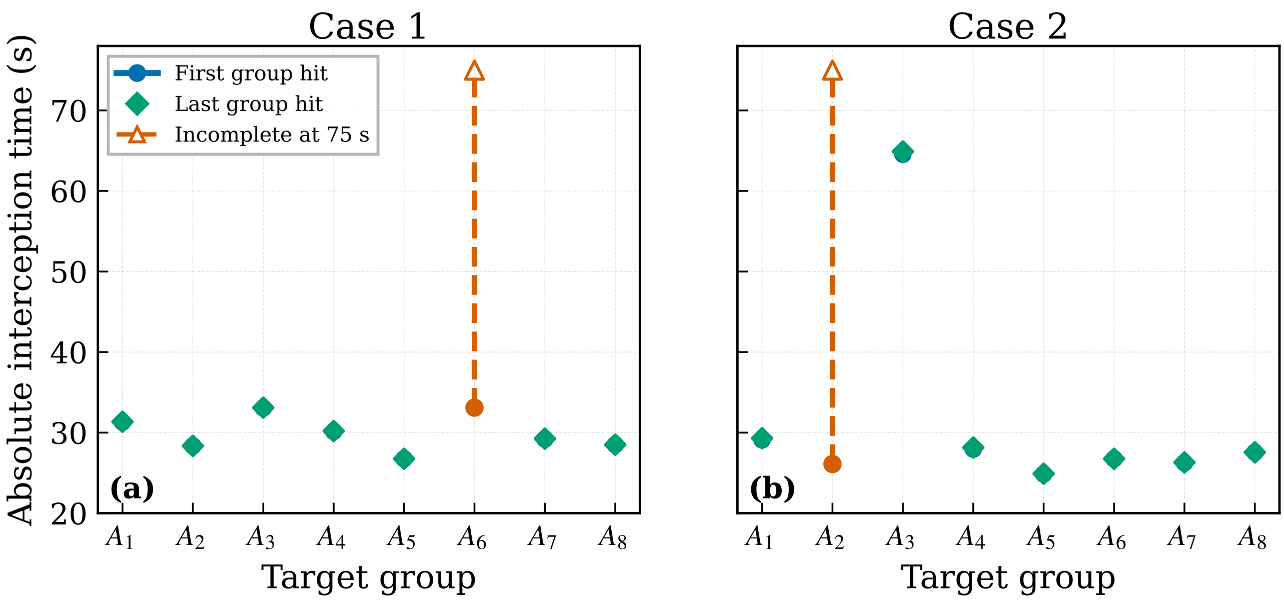

Figure 2: Absolute group interception times. The dashed orange segments mark right-censored incomplete groups (A_6 in Case 1 and A_2 in Case 2). The Case 2 A_3 group completes near 64.9 s, much later than the other complete groups, although its first-to-last hit spread remains within the synchronization tolerance.

Attribution of sensing and actuation effects. To avoid assigning the miss to delay by assumption, we repeated each representative evaluation with the same frozen policy, seed, assignment, and target motion while removing one imperfection at a time. Table 2 reports individual hits H, complete cooperative groups G, and the originally missed pair's minimum distance.

Table 2: Frozen-policy counterfactual diagnosis. All conditions retain 8/8 target coverage. Values are $H/20$, $G/8$, and $d_{\min}$ for the originally missed interceptor (m).

	Case 1			Case 2		
Condition	H	G	$d_{\min}$	H	G	$d_{\min}$
Observed boundary condition	19	7	3.243	19	7	3.574
Remove observation delay only	19	7	3.602	17	6	4.233
Remove measurement noise only	20	8	1.023	20	8	1.078
Ideal observation	20	8	1.714	20	8	1.648
Remove command lag only	19	7	3.432	20	7	2.782
Ideal observation and no lag	20	8	0.725	20	8	1.599

The counterfactual results show two different sensitivity patterns. Case 1 is a noise-dominated narrow-margin failure. Removing measurement noise moves D_6 from 3.243 to 1.023 m and restores all eight groups, whereas removing delay or command lag alone does not recover the hit. Its 0.243 m miss margin is much smaller than the 3 m position-noise standard deviation, and the ideal-observation conditions also restore 20/20 hits. The slight increase of $d_{\min}$ when only delay or lag is removed further confirms that isolated phase changes need not improve a nonlinear terminal encounter.

Case 2 exhibits a nonlinear interaction. Removing measurement noise restores 20/20 hits and 8/8 complete groups, and removing command lag restores the D_2 hit; however, removing delay alone while retaining noisy measurements changes the closed-loop phase and produces 17/20 hits and 6/8 complete groups. The latter result rules out the simplistic interpretation that a 0.05 s observation delay is the sole failure cause or that performance varies monotonically with delay. Instead, the data support a coupled mechanism: delayed/noisy terminal state estimates, command dynamics, high closing speed, and, in Case 2, continuing target maneuver jointly shift the closest-approach geometry. Since this is a nonlinear closed loop, the one-factor removals are diagnostic rather than an additive causal decomposition.

State, control, and cooperative histories. Figs. 3–6 provide the trajectory, acceleration-command, kinematic, attitude, time-to-go, range-to-go, and group-synchronization histories for the same two runs.

Reliability implication and limitation. The results establish a two-level conclusion. At the mission level, redundant assignment prevents a single near-threshold miss from becoming a target penetration: all eight attackers are intercepted in both examples. At the strict group level, the current framework does not explicitly detect closest-approach passage and does

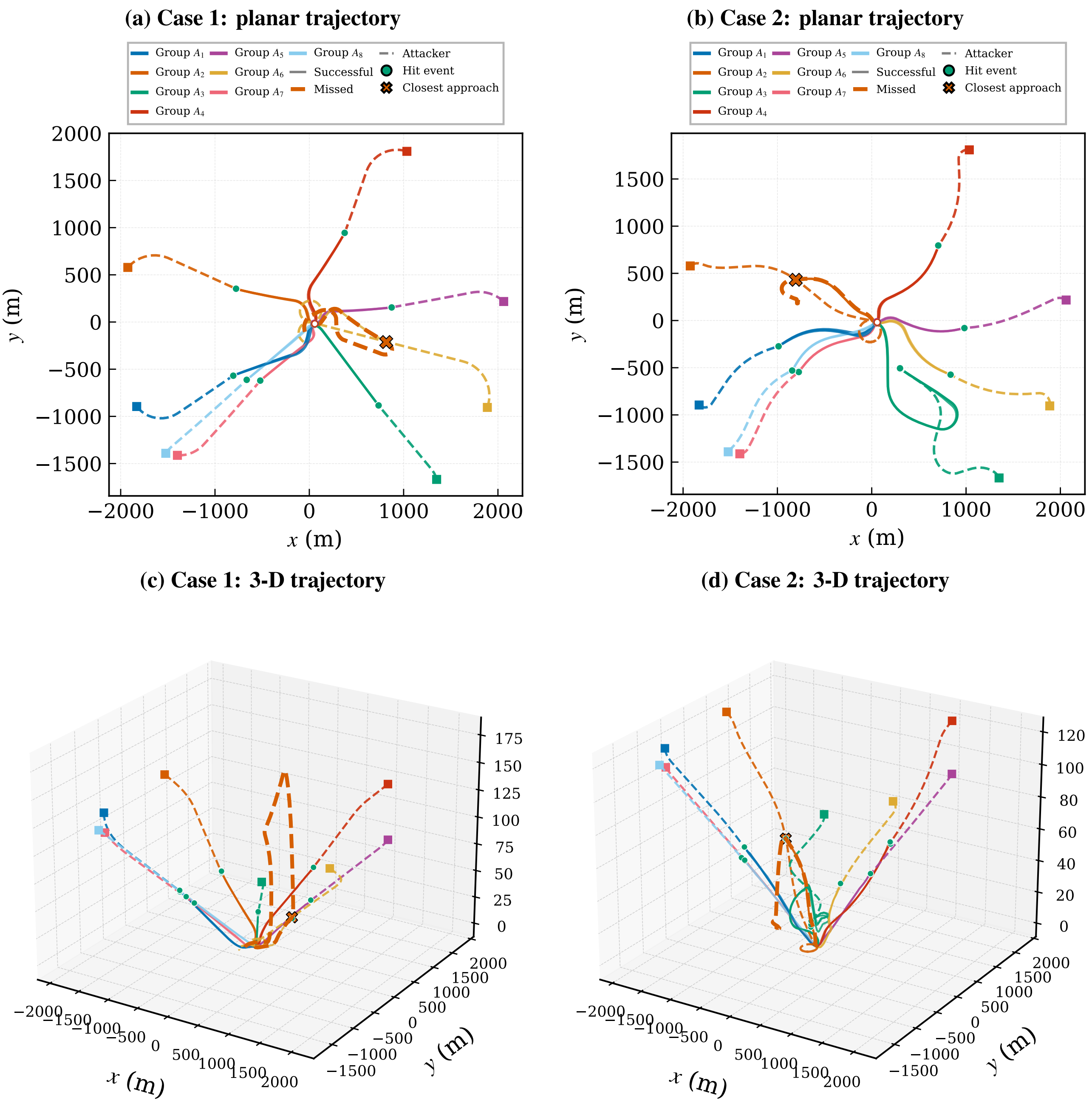

Figure 3: Trajectories for the two boundary failures. The emphasized dashed curve identifies the missed assigned interceptor; circles denote hit events, and the cross denotes its closest-approach point. Redundant assignment preserves 8/8 target coverage despite the individual miss.

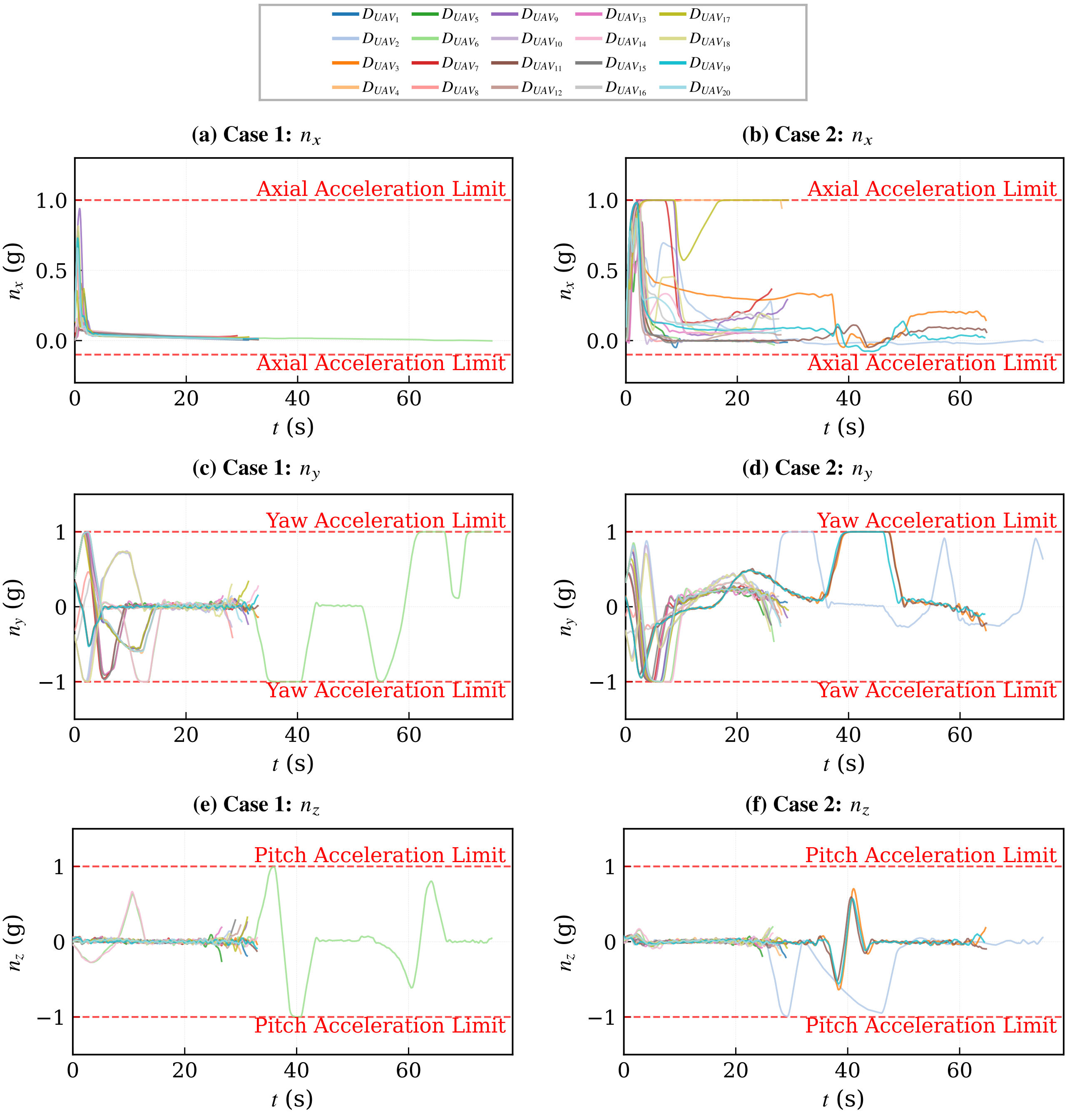

Figure 4: Acceleration-command histories. The missed member receives bounded commands throughout the terminal encounter; its failure is associated with the phase and geometry documented in Fig. 1, not command divergence.

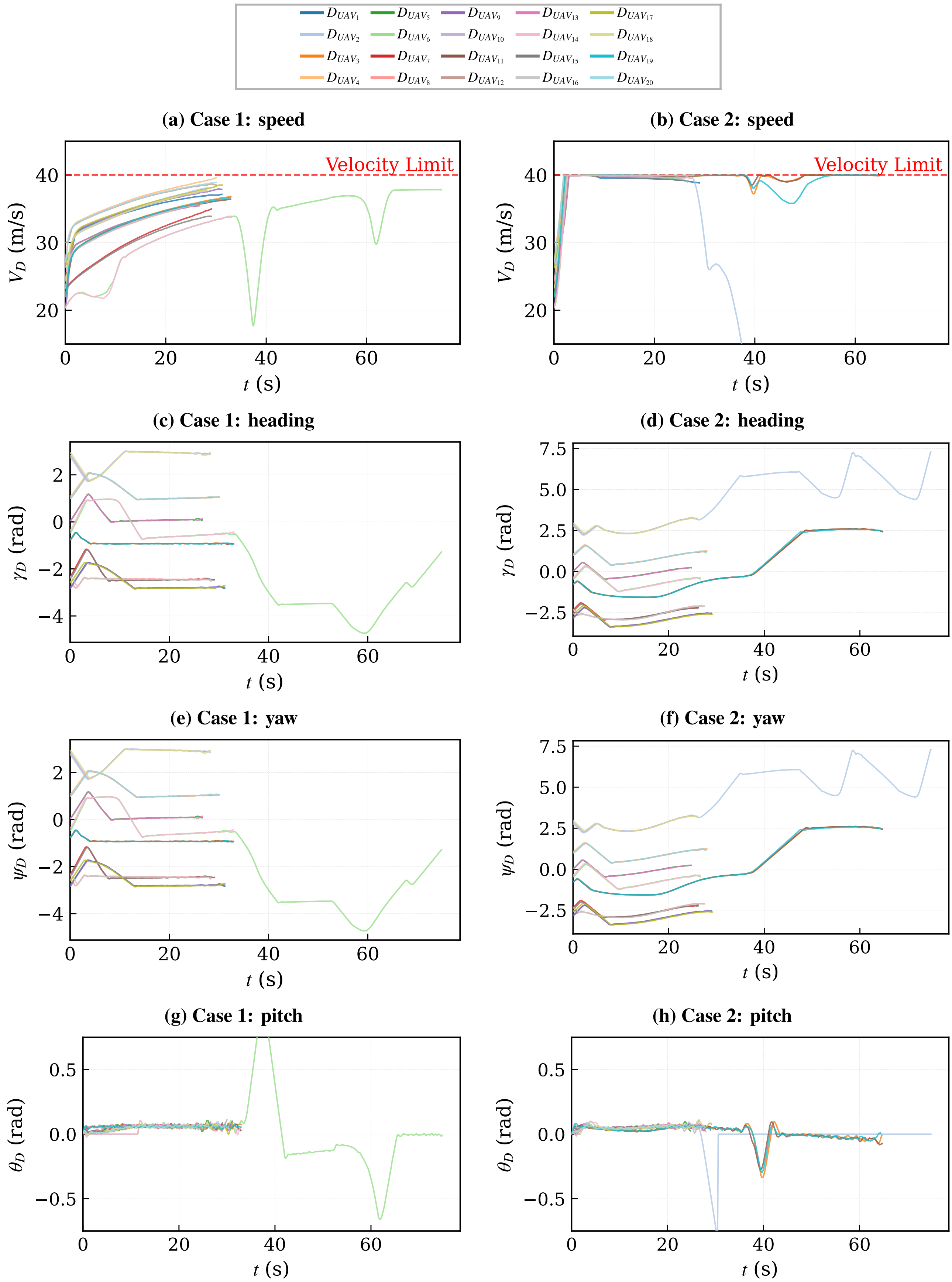

Figure 5: Kinematic and attitude histories. Successful interceptors terminate near their hit instants, whereas the missed member remains active after closest-approach passage.

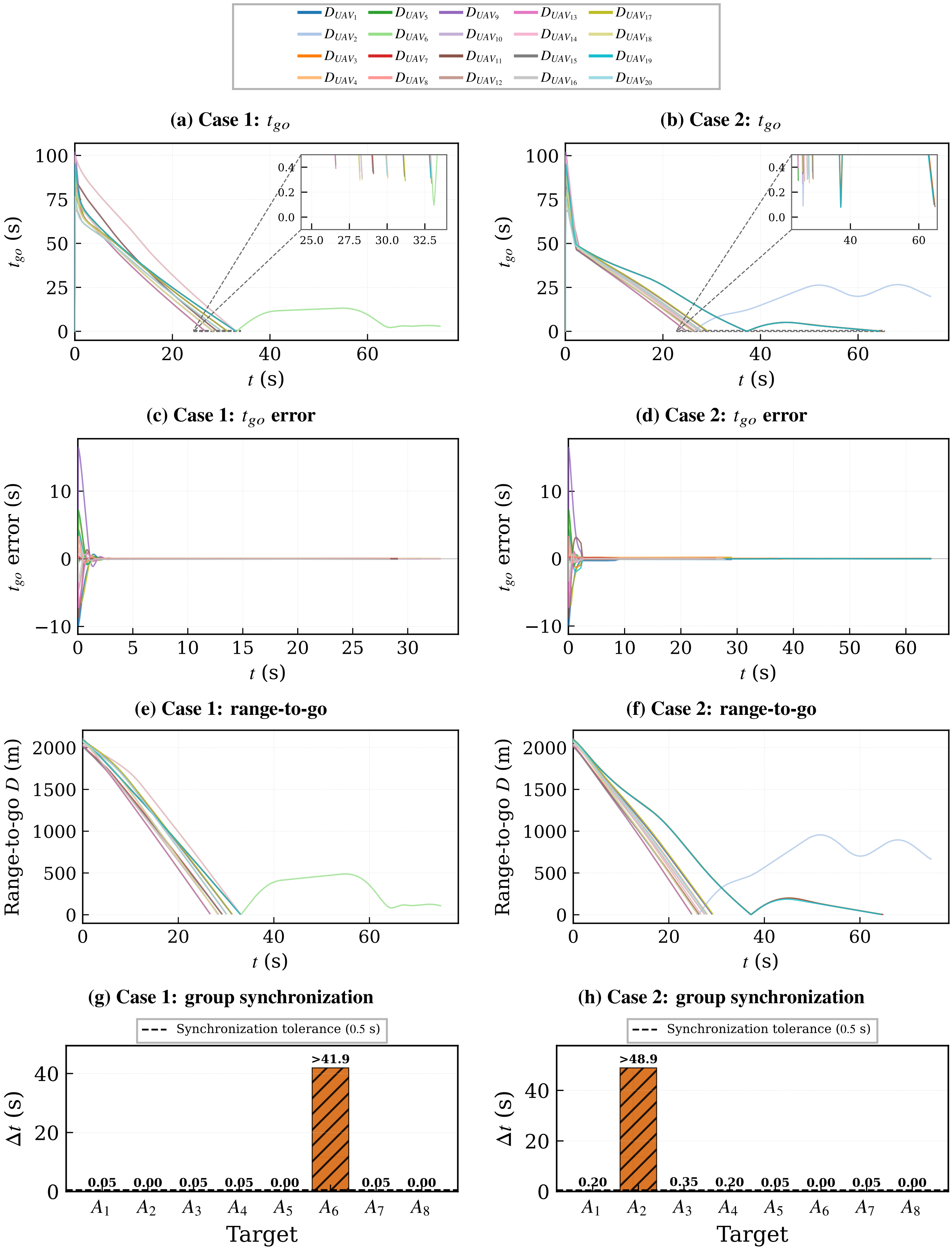

Figure 6: Cooperative-engagement diagnostics. Successful members terminate at interception, whereas the missed member passes its closest point and its range subsequently increases. The hatched A_6 (Case 1) and A_2 (Case 2) bars report the right-censored lower bounds on strict group-completion delay.

not invoke reassignment or a recovery mode after the assigned target has been hit by peer interceptors. It therefore cannot guarantee all-member completion under the coupled terminal uncertainties identified above.

This limitation motivates three concrete extensions: (i) online miss detection using predicted closest-approach distance together with $\dot{d} \geq 0$, (ii) event-triggered target reassignment or reserve interceptor activation, and (iii) uncertainty-aware state estimation and robust training that explicitly cover sensing/actuation phase variations. The added analysis thus characterizes not only where the present framework fails, but also which observable signatures can trigger a practical recovery mechanism.